

Evaluation and Optimization of LLM and RAG Components for a Post-operative Oral Surgery Consultation Chatbot

Sirasit Lochanachit^{1*}, Patcharamon Bunlaue¹, Chanapat Kaewmuneechoke¹, Nopasorn Wilairatanaporn²,
Vorapat Trachoo³, Kritsasith Warin⁴, Praphan Pavarangkoon¹

¹School of Information Technology, King Mongkut's Institute of Technology Ladkrabang,
Bangkok, 10520, Thailand (sirasit@it.kmitl.ac.th) * Corresponding author

²Faculty of Dentistry, Chulalongkorn University,
Bangkok, 10330, Thailand (6430038332@student.chula.ac.th)

³Department of Oral and Maxillofacial Surgery, Faculty of Dentistry, Chulalongkorn University,
Bangkok, 10330, Thailand (vorapat.t@chula.ac.th)

⁴Division of Oral and Maxillofacial Surgery, Faculty of Dentistry, Thammasat University,
Pathum Thani, 12120, Thailand (warin@tu.ac.th)

Abstract: The increasing demand for dental services highlights the need for efficient post-operative oral surgery consultations. Many patients experience anxiety due to limited knowledge of oral care and treatment. This study introduces a chatbot prototype integrating Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) to provide accurate, context-aware responses. The research evaluates various LLMs, embedding models, and chunking techniques to enhance chatbot performance. The multilingual-e5-large embedding model excelled in retrieval tasks due to its multilingual training, instruction tuning, and contrastive pre-training, ensuring high retrieval precision. The Hybrid Chunking method was selected for its ability to segment text contextually, combining Markdown-based, token-based, and semantic segmentation for optimal chunk relevance. The Llama3.3 (70B) model was chosen for its superior fluency, relevance, and ability to handle complex dependencies. The results demonstrate that combining the multilingual-e5-large embedding model, Hybrid Chunking technique, and Llama3.3 (70B) model improves retrieval precision, response accuracy, and relevance, enhancing patient care and operational effectiveness of dental staffs.

Keywords: Post-Operative Oral surgery, Chatbot, Retrieval-Augmented Generation (RAG), Large Language Model (LLM)

1. INTRODUCTION

Patients seeking oral treatments frequently express concerns regarding post-treatment symptoms and self-care requirements. However, dental clinic staff are tasked with responding to repetitive patient inquiries. Although these questions are fundamental and fall within the staff's expertise, addressing them repeatedly imposes a significant workload, thereby reducing the clinic's operational efficiency. While routine dental procedures generally result in minimal post-treatment complications, oral surgeries—such as tooth extractions or impacted wisdom tooth removal—often produce adverse effects, including bleeding, swelling, or infection risks. These conditions necessitate targeted guidance. Effective post-operative care is essential to mitigate complications and accelerate patient recovery; however, patients often lack immediate access to reliable and consistent instructions.

Chatbot technology has become a transformative tool in modern healthcare, offering a scalable solution to improve patient engagement and support. Chatbots are automated systems powered by artificial intelligence (AI) that interact with users in natural language. They provide immediate, consistent, and accessible responses to a wide range of inquiries, thereby reducing the need for direct human intervention. In healthcare settings, chatbots can address common patient concerns, offer self-care advice, and provide crucial post-operative guidance, enhancing

both patient experience and clinic efficiency. Their ability to operate allows patients to access information outside of clinic hours, thus improving their overall satisfaction and reducing the burden on clinic staff. The use of chatbots in healthcare also helps bridge the gap between patients and medical professionals, particularly when access to direct care is limited.

Large Language Models (LLMs) are advanced AI frameworks trained on vast amounts of textual data to understand and generate human-like responses. LLMs, such as OpenAI's GPT models, are particularly effective in processing complex language inputs, enabling them to provide accurate and coherent answers to diverse questions. In healthcare, where precision and clarity are critical, LLMs excel by comprehending domain-specific terminology and adapting to varied conversational contexts. Their integration into chatbot systems significantly improves interaction quality, as they can produce dynamic and context-aware responses that are tailored to individual patient inquiries. This is a marked improvement over traditional rule-based chatbots, which rely on fixed scripts and cannot adjust their responses based on user input.

To address these challenges, this study proposes the development of a chatbot prototype designed for post-operative consultation. The chatbot functions as an interactive media, enabling real-time engagement between pa-

tients and the system by dynamically adapts its responses based on user input, providing personalized and immediate assistance. This capability ensures that patients receive timely and relevant guidance, reducing uncertainty about post-operative care while enhancing their overall experience. However, relying solely on pre-trained knowledge poses limitations, as static models may lack up-to-date or case-specific information.

To enhance the chatbot's efficacy, this prototype incorporates Retrieval-Augmented Generation (RAG) in conjunction with the LLM. RAG is a hybrid approach that augments the LLM's generative capabilities with real-time retrieval of domain-specific information from external sources, such as clinical guidelines or post-surgical care protocols. This integration is crucial as it ensures that responses remain current and tailored to the specialized context of oral surgery, addressing potential limitations in the LLM's static knowledge base. Ultimately, this research contributes a chatbot prototype that facilitates patient access to precise post-operative guidance while optimizing clinic workflows. To support this, we systematically evaluate different embedding models for semantic retrieval quality and compare chunking techniques to determine the most effective strategy for document segmentation in RAG-based systems. These evaluations guide the selection of components that maximize retrieval accuracy and response relevance. By improving patient response and reducing staff workload, the proposed system enhances recovery outcomes and elevates the operational effectiveness of dental practices.

2. RELATED WORK

Post-operative care is a critical aspect of dental treatment, directly influencing patient recovery and overall satisfaction. Inadequate guidance after oral surgery can result in complications such as infections, excessive bleeding, and delayed healing [1]. Traditionally, patients rely on printed instructions or follow-up visits, but these methods may not always be effective in addressing immediate concerns. To bridge this gap, AI-powered chatbots have emerged as a promising tool to enhance patient education and accessibility to dental guidance. Leveraging machine learning and natural language processing (NLP), these chatbots can dynamically generate responses, though they require extensive training data and high computational resources [2].

In dentistry, chatbots have been deployed to assist with general patient education, appointment scheduling, symptom checking, reducing the workload on dental professionals while improving patient satisfaction [3]. However, many existing implementations lack domain-specific knowledge tailored to post-operative oral surgery care, leaving a gap in their ability to provide precise, evidence-based recommendations for patient recovery.

LLMs have significantly enhanced chatbot capabilities by enabling them to generate human-like responses with advanced contextual understanding. Encoder-only

models, such as BERT, specialize in understanding text and are commonly used for classification and information retrieval tasks. Decoder-only models (e.g., GPT-4) excel in text generation, making them effective for open-ended conversations [4]. Encoder-decoder models, such as T5, combine both functions and are widely used for both tasks [5]. Despite their strengths, LLMs pose challenges such as hallucinated responses, inherent biases, and high computational demands [4]. These issues are particularly concerning in healthcare applications, where the accuracy of information is paramount.

RAG addresses these limitations by combining external knowledge retrieval with generative models, improving factual accuracy and contextual relevance [6]. A retrieval module locates relevant information, while a generation module synthesizes coherent responses. This approach enhances chatbot reliability by reducing reliance on the LLM's internal knowledge, which may be outdated or incomplete. However, RAG faces challenges such as retrieval latency and dependency on high-quality knowledge sources [6].

Embedding models play a crucial role in both retrieval and generation processes for RAG-based chatbots. These models convert textual data into numerical vectors, preserving semantic relationships between words and phrases. Contextual embedding models, such as BERT embeddings, generate dynamic word representations based on surrounding text, improving the chatbot's ability to understand meaning in different contexts [6]. The choice of embedding model significantly impacts chatbot performance, as more advanced embeddings enable better retrieval accuracy and response generation.

Text chunking is another critical factor influencing retrieval performance in RAG-based systems. Chunking involves dividing large documents into smaller, manageable text segments for more efficient information retrieval. Recent researches have proposed more adaptive methods, such as Recursive Character Splitting, which recursively breaks text based on character length while preserving sentence boundaries, and hybrid approaches that combine structural, token-based, and semantic cues to generate more contextually relevant chunks [7], [8].

The Dental Loop Chatbot [9] represents a notable application of RAG-based AI in dentistry. This chatbot provides real-time, evidence-based guidance for dental practitioners by retrieving and synthesizing information from clinical guidelines and academic literature. While the Dental Loop Chatbot primarily assists dental professionals in clinical decision-making, our project extends this application to post-operative oral surgery care for patients. By shifting the focus from practitioner support to direct patient guidance, our chatbot aims to address common concerns such as bleeding control, swelling management, and infection prevention. This patient-centered approach differentiates our work from existing implementations and highlights the potential for AI-driven solutions in improving post-surgical recovery.

3. METHODOLOGY

3.1 System Architecture Overview

The chatbot's development follows a structured workflow that leverages RAG to enhance response quality by integrating retrieval from a knowledge base with LLM generation. The system architecture is designed to process user queries efficiently and generate contextually relevant responses through three key stages: Data Processing, Query Processing and Response Generation as illustrated in Fig. 1.

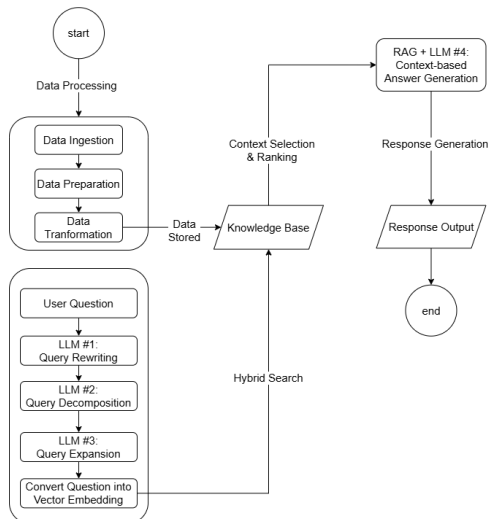


Fig. 1. System Architecture Overview

3.2 Data Processing

The knowledge base is constructed through a systematic process starting with data collection from 34 PDF files of clinical practice guidelines. The data undergoes cleaning, including normalization of special characters and spacing, followed by document chunking to divide the text into manageable segments. Metadata is then added to each chunk to aid retrieval. Next, the chunks are transformed into vector embeddings using embedding model. Finally, the vectors are stored in a knowledge base with a hybrid retrieval system, combining keyword-based and semantic search, and optimized through clustering for efficient retrieval.

3.3 Query Processing

User queries are refined through a series of LLMs to improve clarity and specificity. This involves rewriting, expanding, and decomposing queries into sub-queries, which are then converted into vector embeddings. This step ensures the system understands user intent accurately before retrieving relevant context.

3.4 Response Generation

The system generates accurate, contextually relevant responses by integrating the knowledge base with a LLM using RAG. Retrieved context is combined with the user query, enhancing answer quality through RAG and LLM synergy.

4. EXPERIMENTAL DESIGN AND EVALUATION

To develop our chatbot post-operative oral surgery consultation, we conducted three experiments to optimize its key components: Embedding Model, Chunking Technique, and LLM. The first two experiments are evaluated using three evaluation methods; Generation-based Evaluation, Retrieval-based Evaluation, and LLM-as-a-Judge. The third experiment used the same three evaluation methods and additionally performed Human Evaluation.

4.1 Evaluation Method

To comprehensively assess the system-generated responses, four key evaluation methods were utilized.

4.1.1 LLM-as-a-Judge

An LLM-as-a-Judge approach was adopted to assess the quality of the system-generated responses. The LLM was prompted to evaluate the responses based on the three key criteria: Correctness, Relevance, and Helpfulness. A Likert scale from 1 to 5 was utilized, where 1 represents "strongly disagree" and 5 represents "strongly agree". The scores were categorized as follows: scores of 1 and 2 were classified as "Bad", a score of 3 was classified as "Neutral", and scores of 4 and 5 were classified as "Good".

4.1.2 Generation-based Evaluation

Generation-based Evaluation measures the quality of system-generated responses by comparing with the golden responses provided by domain experts using four key metrics. BERTScore evaluates semantic similarity, BLEU assesses n-gram precision, METEOR considers synonyms and paraphrasing for completeness, and ROUGE-I focuses on recall to ensure information coverage. These metrics provide a comprehensive assessment of response accuracy and relevance.

4.1.3 Retrieval-based Evaluation

Retrieval-based Evaluation assesses the performance of the RAG system by focusing on the effectiveness of its retrieval mechanism, utilizing eight key metrics. Precision@K measures the accuracy of retrieved results, while Recall@K evaluates the proportion of relevant items captured. MRR and nDCG assess ranking quality, with MRR focusing on the first relevant item and nDCG accounting for relevance and position. Similarity Score gauges semantic alignment between retrieved passages and the query, Groundedness Score ensures factual consistency, Completeness Score evaluates the comprehensiveness of retrieved information, and Relevance Score determines the overall pertinence of retrieved documents. These metrics collectively ensure the quality and accuracy of the retrieval process.

4.1.4 Human Evaluation

The evaluation was conducted by 10 dentists, who were tasked with assessing the system-generated re-

sponses based on the same three criteria used in the LLM-as-a-Judge: Correctness, Relevance, and Helpfulness.

4.2 Component Optimization Experiments

To optimize our chatbot for post-operative oral surgery consultation, we conducted three experiments to refine its components.

4.2.1 Embedding Model Comparison

Three embedding models: bge-m3, gte-Qwen2 (1.5B), and multilingual-e5-large were evaluated to determine their effectiveness in enhancing response accuracy and relevance. During the evaluation, the Llama3.3 (70B) model and Recursive Character Splitting chunking method were held constant. To assess output quality, the LLM-as-a-Judge was DeepSeek-R1-Distill-Llama (70B).

4.2.2 Chunking Technique Comparison

Two chunking techniques: Recursive Character Splitting and Hybrid Chunking were compared to analyze their impact on retrieval efficiency and response quality. The top-performing embedding model from the embedding model evaluation, along with the Llama3.3 (70B) model, was utilized in this experiment. For LLM-as-a-Judge evaluation, the same DeepSeek-R1-Distill-Llama (70B) model was employed to provide consistent judgment across experiments involving embedding and chunking components.

4.2.3 Language Model Comparison

Two LLMs: Llama3.3 (70B) and DeepSeek-R1-Distill-Llama (70B)—were assessed to identify which model produces the most accurate and contextually appropriate responses. The best embedding model and chunking technique identified from the previous experiments were applied in this evaluation. In this experiment, a reciprocal evaluation was conducted: Llama3.3 (70B) was used to judge responses generated by DeepSeek-R1, while DeepSeek-R1 served as the judge for responses generated by Llama3.3 (70B). This mutual assessment approach enabled cross-model comparison from different evaluative perspectives.

5. RESULT

This section presents the results of the study, focusing on three experiments: Embedding Model Evaluation, Chunking Technique Evaluation, and LLM Evaluation. The first two experiments were conducted using three evaluation methods: LLM-as-a-Judge, Generation-based Evaluation, and Retrieval-based Evaluation to assess the system's performance. The same three evaluation methods were performed on the third experiments and includes Human Evaluation. To ensure robust and generalizable performance evaluation, all experiments were conducted using 16 frequently asked user queries in Thai, representative of common post-operative patient concerns, with golden responses provided by dentists as domain experts.

5.1 Experiment 1: Embedding Model Comparison

To identify the most suitable embedding model for enhancing retrieval performance in the proposed consultation system, 3 embedding models: **bge-m3**, **gte-Qwen2 (1.5B)**, and **multilingual-e5-large** were evaluated on 16 queries using three methods: LLM-as-a-Judge, Generation-based Evaluation, and Retrieval-based Evaluation.

Through **LLM-as-a-Judge** evaluation, multilingual-e5-large outperformed all other models across every criterion (Table 1). This indicates that from the perspective of LLM, its embeddings resulted in responses that were not only accurate but also highly relevant and helpful. In contrast, gte-Qwen2 and bge-m3 had slightly lower but still reasonable performance.

In the **Generation-based Evaluation**, automatic evaluation metrics were used to compare generated responses with golden responses. All models scored zero on BLEU due to paraphrasing in retrieved answers, highlighting the limitation of lexical overlap metrics. However, multilingual-e5-large achieved the highest overall average of 0.25, closely followed by gte-Qwen2 with 0.24 and bge-m3 with 0.23. BERTScore values were comparable across models, indicating similar semantic alignment with reference answers.

Table 1. Three Evaluation Metrics for Embedding Models

Evaluation	Metric	bge-m3	Qwen2	multi-e5
LLM Judge (Score: 0-5)	Correctness	4.2	4.4	4.7
	Relevance	4.4	4.8	4.9
	Helpfulness	4.3	4.4	4.7
	Average	4.3	4.5	4.8
Generation (Score: 0-1)	BERTScore	0.72	0.74	0.72
	BLEU	0	0	0
	METEOR	0.02	0.05	0.04
	ROUGE-I	0.16	0.15	0.22
	Average	0.23	0.24	0.25
Retrieval (Score: 0-1)	Precision@3	0.40	0.40	1
	Recall@3	0.56	0.56	1
	MRR	0.53	0.46	1
	nDCG	0.52	0.49	1
	Similarity	0.80	0.60	0.92
	Groundedness	0.65	0.65	0.88
	Completeness	0.62	0.61	0.86
	Relevance	0.74	0.65	0.89
	Average	0.60	0.55	0.94

Lastly, the **Retrieval-based Evaluation** revealed the most significant performance gap. Multilingual-e5-large achieved a near-perfect average score, with full marks in Precision@3, Recall@3, MRR, and nDCG, along with leading scores in Similarity, Groundedness, Completeness, and Relevance. bge-m3 slightly outperformed gte-Qwen2 in certain aspects such as MRR and Similarity, but both were considerably weaker in precision and overall retrieval quality.

Across three evaluation methods, multilingual-e5-

large proved to be optimal, excelling in Retrieval-based Evaluation. Its multilingual training, instruction tuning, and encoder-only architecture enhance retrieval precision and relevance, making it the most suitable choice for the next experiment.

5.2 Experiment 2: Chunking Technique Comparison

To compare the effectiveness of Recursive Splitting and Hybrid Chunking strategies, we conducted a comprehensive evaluation using three perspectives: LLM-as-a-Judge, Generation-based Metrics, and Retrieval-based Metrics. Averaging on 16 queries, the results varied across evaluation types, reflecting different strengths of each approach.

LLM-as-a-Judge evaluation reveals that Recursive Splitting received notably higher scores, possibly due to its preference for the larger, more variably sized chunks which retained broader context (Table 2).

Table 2. Three Evaluation Metrics for Chunking Techniques

Evaluation	Metric	Recursive Splitting	Hybrid Chunking
LLM Judge (Score: 0-5)	Correctness	4.7	4.19
	Relevance	4.9	4.38
	Helpfulness	4.7	4.19
	Average	4.8	4.25
Generation (Score: 0-1)	BERTScore	0.72	0.75
	BLEU	0	0
	METEOR	0.04	0.04
	ROUGE-I	0.22	0.20
	Average	0.25	0.25
Retrieval (Score: 0-1)	Precision@3	1	1
	Recall@3	1	1
	MRR	1	1
	nDCG	1	1
	Similarity	0.92	0.92
	Groundedness	0.88	0.91
	Completeness	0.86	0.90
	Relevance	0.89	0.89
	Average	0.94	0.95

In the **Generation-based Evaluation**, both methods showed similar performance. Hybrid Chunking slightly outperformed in BERTScore, reflecting better semantic similarity to references. On the other hand, Recursive Splitting performed marginally better in ROUGE-I. However, both methods scored 0 on BLEU and similarly low on METEOR, indicating limited n-gram overlap with reference answers.

Lastly, the **Retrieval-based Evaluation** revealed only a slight advantage for Hybrid Chunking. Both techniques achieved perfect scores in Precision@3, Recall@3, MRR, and nDCG, highlighting their strong retrieval capabilities. However, Hybrid Chunking showed better performance in Groundedness and Completeness, pointing to more contextually complete and well-grounded responses.

In summary, the LLM judge favored Recursive Splitting's larger, variable chunks for their extensive context, but Hybrid Chunking's uniform chunks better suit RAG systems by enhancing retrieval efficiency, preserving query alignment, and delivering comprehensive responses. Its consistency aligns with retrieval demands, making it the superior choice, adopted for Experiment 3.

5.3 Experiment 3: LLM Comparison

Two LLMs—Llama3.3 (70B) and DeepSeek-R1-Distill-Llama (70B)—were evaluated on the same 16 queries across four methods: Human, LLM-as-a-Judge, Generation-based, and Retrieval-based.

In the **Human Evaluation**, conducted by a panel of 10 dentists, Llama3.3 (70B) emerged as the superior model as depicted in Fig. 2. These results suggest its strong capabilities in delivering accurate and contextually appropriate responses, due to long-context understanding. DeepSeek-R1's distillation approach, while efficient, led to lower scores, particularly in Helpfulness, likely from reduced detail and nuance.

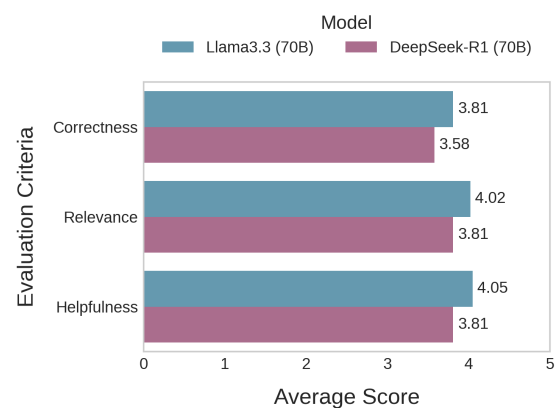


Fig. 2. Human Evaluation Results by Dentists (n=10)

On the contrary, the **LLM-as-a-Judge** evaluation favored DeepSeek-R1-Distill-Llama (70B), which scored higher across all criteria as shown in Fig. 3. These results may reflect a judge model's preference for concise and well-aligned outputs, which are characteristics often enhanced by the distillation process.

The **Generation-based Evaluation** showed a slight advantage for Llama3.3 (70B), outperforming in BERT Score, METEOR, and ROUGE-I, implying that (Table 3) Llama3.3 (70B) generated outputs that were closer in semantic similarity and fluency.

In the **Retrieval-based Evaluation**, both models demonstrated strong performance with equal average scores, and perfect marks across Precision@3, Recall@3, MRR, and nDCG. However, Llama3.3 (70B) slightly edged out in Similarity and Groundedness, reflecting its stronger contextual alignment and integration of retrieved knowledge effectiveness for consultations.

Fig. 4 summarizes overall performance across all four evaluation methodologies. Llama3.3 (70B) outperformed DeepSeek-R1-Distill-Llama (70B) in Human and

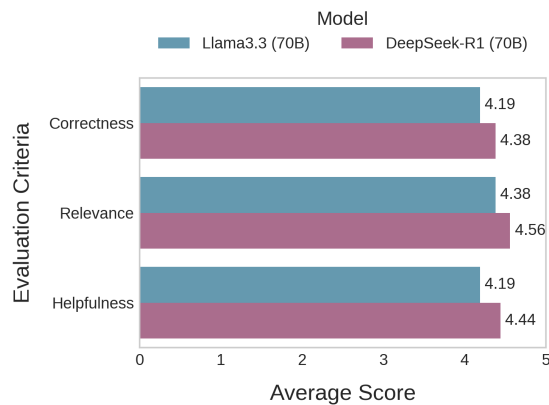


Fig. 3. LLM-as-a-Judge Results

Table 3. Generation and Retrieval Metrics for 70B LLMs

Evaluation	Metric	Llama3.3	DeepSeek-R1
Generation (Score: 0-1)	BERTScore	0.75	0.72
	BLEU	0	0
	METEOR	0.04	0.03
	ROUGE-I	0.20	0.12
	Average	0.25	0.22
Retrieval (Score: 0-1)	Precision@3	1	1
	Recall@3	1	1
	MRR	1	1
	nDCG	1	1
	Similarity	0.92	0.90
	Groundedness	0.91	0.90
	Completeness	0.90	0.88
	Relevance	0.89	0.89
	Average	0.95	0.95

Generation-based evaluations due to its advanced fine-tuning and long-context understanding, delivering detailed, accurate responses. DeepSeek excelled in LLM-as-a-Judge with concise outputs favored by the judge, but Llama’s edge in relevance and groundedness across most metrics makes it the preferred choice.

6. CONCLUSION

This study optimized a chatbot prototype for post-operative oral surgery consultations using RAG and LLMs by evaluating key components. The multilingual-e5-large embedding model demonstrated superior retrieval accuracy due to its multilingual and instruction-tuned capabilities. Hybrid Chunking outperformed other method by improving indexing and response completeness, crucial for precise medical information. Among LLMs, Llama3.3 (70B) offered the most contextually accurate and fluent responses. Multilingual-e5-large, Hybrid Chunking, and Llama3.3 (70B) were selected for their combined strengths in accuracy, precision, and relevance, optimizing the chatbot’s performance for post-operative oral surgery consultations.

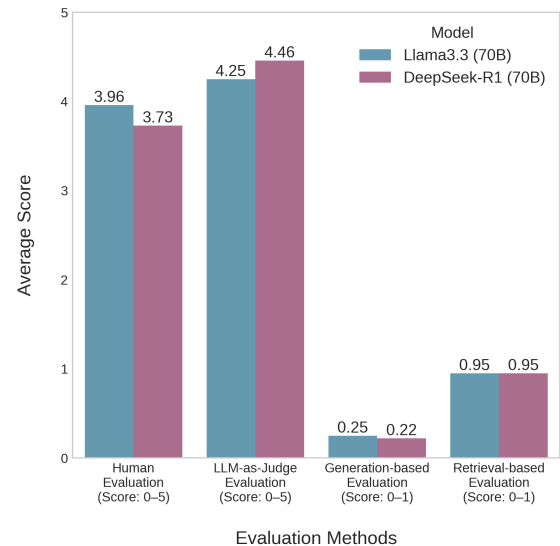


Fig. 4. Overall performance summary across all four evaluation methodologies

REFERENCES

- [1] O. Ayo-Farai et al., “Postoperative Infections: Risk Factors, Prevention, and Management Strategies” *International Journal of Medical Science and Dental Research*, vol. 7, no. 4, pp. 52-76, 2024.
- [2] Gökçearsan, Şahin, et al. “Benefits, Challenges, and Methods of Artificial Intelligence (AI)-Supported Chatbots in Education: Systematic Literature Review.” *Contemporary Educational Technology*, vol. 15, no. 2, 2023.
- [3] Z. Othman et al., “AI-powered dental consultation chatbot: Enhancing public healthcare accessibility and awareness,” *Journal of Advanced Research in Applied Sciences and Engineering Technology*, vol. 57, no. 2, pp. 14-27, 2024.
- [4] Brown, Tom, et al. “Language Models Are Few-Shot Learners.” *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877 - 1901, 2020.
- [5] Raffel, Colin, et al. “Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer.” *Journal of Machine Learning Research*, vol. 21, pp. 1-67, 2020.
- [6] Gao, Yunfan, et al. “Retrieval-Augmented Generation for Large Language Models: A Survey.” *arXiv preprint arXiv:2312.10997*, 2023.
- [7] Narimissa, Esmaeil, and David Raithel. “Exploring Information Retrieval Landscapes: An Investigation of a Novel Evaluation Techniques and Comparative Document Splitting Methods.” *arXiv preprint arXiv:2409.08479*, 2024.
- [8] Zhao, Jihao, et al. “MoC: Mixtures of Text Chunking Learners for Retrieval-Augmented Generation System.” *arXiv preprint arXiv:2503.09600*, 2025.
- [9] M. S. H. Arian et al., “Dental Loop Chatbot: A prototype large language model framework for dentistry,” *Software*, vol. 3, no. 4, pp. 587-594, 2024.